

Artem Yushkovskiy

AI Systems Engineer & Tech Lead • Agentic AI & Distributed Systems • Cloud-native (K8s, GCP, AWS)

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Summary

AI Systems Engineer & Tech Lead, 8+ years building distributed AI platforms and agentic infrastructure (10+ in software engineering); tech lead of a 10-person squad (no direct reports) owning architecture, mentoring, and delivery.

Creator of asya.sh, an open-source Kubernetes-native actor mesh for AI/agentic workloads, presented at **KubeCon EU 2026** applied it in production for **self-hosted GenAI image pipelines** (100K+ images/night), **per-country fraud-detection models** across 20+ countries (<10ms, 70% fraud reduction), and an **internal ML platform** on GCP Vertex AI used by 5+ teams.

Berlin-based, authorized to work in Germany (no visa sponsorship needed) — seeking hybrid or fully remote full-time roles in **agentic AI / distributed platform engineering** (permanent employment, not freelance).

Experience

Sr AI Engineer - Distributed AI Orchestration, GenAI (images), Delivery Hero – Berlin, DE

Feb 2024 – present

- Owned architecture decisions for a 10-person squad (5 engineers, 5 data scientists); championed modular AI over monolithic models: smart routing to specialized models per problem type rather than one large black-box model, enabling divide-and-conquer at scale with clear ownership boundaries (engineers own infra/pre-processing, data scientists own model logic and deployment via platform tooling)
- Designed and scaled a self-hosted image-enhancement platform for dish & menu imagery (upscaling, outpainting, background removal, color balance) on AWS/Kubernetes; GPU-bound multi-model pipelines per image (production models, scoring, analysis, guardrails) under strict cost constraints; 100K+ images nightly in batch, migrated to near-realtime streaming (<5 min SLA, shrinking) with zero application code changes; early A/B tests show measurable lift in clicks and purchases at global scale
- Built a data-centric ML quality-control system to close the image-enhancement feedback loop; designed a human labeling & exploration tool over a 223K-image dataset (ordinal severity + multi-label defect taxonomy), aligned a team-reviewed golden set and scaled it 1K → 20K via VLM labeling, then trained an ordinal ConvNeXt classifier (0.87 severe-case recall on the human test set; VLM-val vs human-test macro-F1 gap <0.005) with one-click weekly retraining — making consistent, human-aligned data (not model architecture) the driver of quality [PyTorch, MLflow]
- Designed CI/CD for a monorepo of 15+ model packages building multi-GB GPU images; cut end-to-end deployment from days to ~1 hour and image sizes 3-4x smaller (optimized layer caching cut build & pod-startup times ~5-6x) [Docker Buildx, uv]
- Recognized the actor mesh pattern's general value for AI and agentic workloads beyond image processing; drove the initiative to re-implement and open-source it as asya.sh (Crossplane-based operator, CRDs, KEDA autoscaling, pluggable transports); now live and adopted internally
- Authored company-wide RFC on GPU capacity and cost strategy (training + serving): 18-36 month demand forecasting, usage patterns across teams, negotiation baseline for cloud provider agreements; defined SLA targets and cost governance for GPU workloads
- Mentored engineers through two promotions (junior → mid, mid → senior); continuously coaching data scientists on engineering best practices (testing, CI/CD, clean code) to raise tech culture across the team; active contributor to ML chapter and weekly operations reviews, driving cross-team knowledge sharing

ML Engineer, ML Platform → Fraud Detection → GenAI, Delivery Hero – Berlin, DE

Nov 2021 – Feb 2024

- Designed and built an internal ML Platform on GCP Vertex AI, providing shared data and ML pipeline infrastructure across teams (CLV, food ontology, dish normalization, email suspicion, fraud detection)
- Owned the full lifecycle of 20+ per-country voucher-fraud models across 2 global regions (EMEA, APAC): built end-to-end CI/CD enabling data scientists to iterate fast against rapidly shifting fraud patterns (nightly batch pipelines on BigQuery/Airflow, per-country training on Vertex AI, automated evaluation with drift tracking via Evidently, results reported to GitHub Actions for promotion decisions); cut voucher fraud 70% (~\$100K+/month saved)
- Two-model serving pipeline: email-suspicion score (DynamoDB caching) plus near-realtime features → CatBoost fraud classifier → rule-based system; FastAPI, optimized to <10ms per request on every food order
- Engineered dataset versioning for strict model comparability without data leakage; re-implemented across two teams and proposed as a company-wide standard
- On-call SRE for fraud models and global Customer Data Platform (low-latency feature aggregation, Flink, DynamoDB); monitoring and observability (DataDog, OpsGenie)
- Co-authored with Google: "[How Delivery Hero connected GitHub with Vertex AI to manage 20+ voucher fraud detection models](#)" (Google Cloud Blog, Nov 2023)
- Transitioned into GenAI: built and maintained early image-generation pipelines (GCP Vertex AI, Pub/Sub, Cloud Run, OpenAI, Imagen), laying the groundwork for the self-hosted platform built in the following role

Software Engineer → MLOps Engineer, Neuromation / Neu.ro – St Petersburg, RU 2018 – 2021

- Built core of a multi-cloud Kubernetes-based ML platform ("Docker-like experience on any cloud and bare metal"): REST API microservices, CLI/SDK, user management, distributed storage, resource orchestration
- Prototyped and integrated ML tooling into the platform (DVC, Seldon, Feast, Pachyderm); ran internal Kubernetes seminars for the team
- Led production computer vision projects for clients (object detection, classification, RabbitMQ-driven inference, MongoDB, Kubernetes)

Security Researcher, Application Security Research 2014 – 2016

- Automated QA of web application security scanners and firewalls (DAST/SAST/WAF); vulnerability search, static code analysis, abstract interpretation, SMT solvers

Projects

Asya 🦄 - Open-source Kubernetes-native Actor Mesh for AI Orchestration Nov 2025

- Distilled from 3+ years of production AI workloads; Crossplane-based, sidecar architecture (Python runtime + Asya sidecar), pluggable transports (SQS, RabbitMQ, Pub/Sub), pluggable state backends (S3, GCS, Redis, NATS KV)
- Stateless actors with dynamic routing ("the message knows the way"); flow compiler translates Python control flow into actor graphs; KEDA-based per-actor autoscaling including scale-to-zero
- Gateway protocols: A2A agents, MCP tools, SSE streaming, REST; pure Python handlers with zero framework coupling, testable locally
- Clean separation of business logic (data scientists) and infrastructure config (platform engineers), solving the ownership handover problem in AI teams ("two files, two owners")
- asya.sh | [GitHub](#)

Public Speaking

- [KubeCon EU 2026](#) Amsterdam (13k+), [AI Infra Summit](#) Munich (500+), [AI in Production](#) Berlin (150+), [Agents in Production](#) virtual
- MLOps Community Berlin meetup organizer; led Ask-Me-Anything initiative in MLOps Community Slack (2021-2023)

Education

Aalto University (Helsinki) & ITMO University (St Petersburg), MSc in Computer Science 2016 – 2018

- Double-degree programme: Information Security and Cloud Computing, with honors.
- *Thesis*: Automated Analysis of Weak Memory Models.

ITMO University (St Petersburg), BSc in Computer Science 2012 – 2016

- Information Security, with honors.
- *Thesis*: Development of the Code Property Graph Construction Module for the Static Analyser ApplicationInspector.Net

Skills

AI Infrastructure & Inference: self-hosted model serving, inference optimization, GPU orchestration (KEDA, scale-to-zero), batch & streaming inference, async actor-based pipelines, AI guardrails, image generation (SDXL), LLM serving (vLLM), PyTorch

Platform Engineering & Developer Experience: Kubernetes (CRDs, operators, Helm, Kustomize), GitOps (ArgoCD, Flux), IaC (Terraform), CI/CD (GitHub Actions, Docker Buildx), internal developer platform, multi-tenancy, RBAC, cost optimization, observability (Prometheus, Grafana, DataDog), secret management (Vault)

MLOps & Data: ML lifecycle (Vertex AI, MLflow), pipeline orchestration (Airflow, KFP), feature stores, dataset versioning (DVC), A/B testing, experimentation, data quality (Evidently), vector search (Redis), stream processing (Flink), low-latency model serving (<10ms)

Programming & Cloud Infrastructure: Python (asyncio, FastAPI, uv), Bash, Linux, AWS (EKS, SQS, SNS, Lambda, ECR), GCP (Vertex AI, Pub/Sub, Cloud Run, Cloud Build, GCS, Artifact Registry), PostgreSQL, DynamoDB, Redis, RabbitMQ, NATS, MongoDB

Spoken Languages: English, Russian (fluent); French (intermediate); German (learning)

Certifications: [Google Cloud Professional Cloud Architect](#) (2025), [Google Cloud Professional Machine Learning Engineer](#) (2023, expired)